

Payments and Transfer Services

Reference for payment lifecycles and failure handling.

Internal Transfers

An internal transfer moves funds between accounts within the same bank and can normally be atomic when both accounts are in one transactional boundary.

Both sides of the movement must succeed or neither should commit.

Inter-Bank Transfers

An inter-bank transfer involves a sending institution, a payment/clearing network and a receiving institution.

Typical lifecycle: initiation, validation, authorization, source debit or reservation, network submission, routing, destination processing, acknowledgement, settlement and reconciliation.

BankFlow uses a simulated payment network and does not connect to actual NPCI infrastructure.

Payment Statuses

INITIATED: request created.

VALIDATING: request checks are running.

DEBITED: source posting completed.

SENT: payment submitted to network.

CREDIT_PENDING: destination outcome not yet confirmed.

CREDITED: destination posting confirmed.

COMPLETED: workflow completed.

FAILED: payment could not complete.

REVERSAL_PENDING: compensation required.

REVERSED: source effect compensated.

RECONCILIATION_REQUIRED: outcome is ambiguous and needs investigation.

Failure Scenarios

Network timeout can leave remote state uncertain.

Duplicate requests require idempotency keys.

Destination unavailability may require retry, reconciliation or reversal.

Acknowledgement loss can occur after the receiving side has actually succeeded.

Payment Rails — Conceptual

Instant payment systems prioritize rapid account-to-account transfer.

NEFT-like systems can be modeled as electronic transfers with defined processing/settlement behavior.

RTGS-like systems model real-time gross settlement concepts for eligible transactions.

BankFlow models concepts only and does not reproduce production protocols.

Beneficiaries

Production systems may apply verification, cooling periods, transfer limits and additional authentication. BankFlow can simulate these controls.